FORMATIVE ASSESSMENT – 2 (FA-2)

MATHEMATICS • CLASS 9 • GENERATING NEW PROBLEMS

Problem 1

Two supplementary angles are in the ratio **7 : 5**. Find the measure of each angle. Formulate a similar problem where two complementary angles are in the ratio **4 : 5** and verify your answer.

Problem 2

In a given figure, lines **AB** and **CD** intersect at point **O**. If **∠AOC + ∠BOE = 70°** and **∠BOD = 40°**, find **∠BOE** and reflex **∠COE**.

Problem 3

If two parallel lines are intersected by a transversal line such that a pair of alternate interior angles are given by **(3x − 15)°** and **(2x + 25)°**, determine the numerical value of **x** and all eight angles formed.

Problem 4

Write the coordinates of the vertices of a rectangle whose length is **6 units** and breadth is **4 units**, such that one vertex lies at the origin **(0, 0)**, the longer side lies along the positive x-axis, and one vertex lies in the first quadrant.

Problem 5

Plot the points **P(−2, 4)**, **Q(3, 4)**, **R(3, −1)**, and **S(−2, −1)** on a Cartesian plane. Identify the geometric shape formed by joining them in order and calculate its area.